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Swaragram: A software toolbox for musical feature of Indian music



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ABSTRACT

We present a melodic musical feature and a toolbox preferable for Indian classical music that incorporates tonic information and micro-tonic scales (Shruti). The Chromagram feature, widely used for computational analysis in Western music, inspired our feature's methodology. The feature's performance is evaluated for Raag Identification and is comparable to other approaches on the same dataset. We also present a few applications of the toolbox.

Code metadata

Current code version	*v1.0*
Permanent link to code/repository used for this code version	https://github.com/SoftwareImpacts/SIMPAC-2022-295
Permanent link to Reproducible Capsule	https://codeocean.com/capsule/6439633/tree/v1
Legal Code License	*GPL-3.0 license*
Code versioning system used	*Git*
Software code languages, tools, and services used	*Python*
Compilation requirements, operating environments & dependencies	*Dependencies on Python libraries: essentia, audioread, librosa, numba>=0.48.0, numpy, PySoundFile, scipy*
If available Link to developer documentation/manual	https://github.com/yeshwantsingh/swaragram/blob/master/core.md, https://github.com/yeshwantsingh/swaragram/blob/master/utils.md
Support email for questions	yeshwantsinghbit@gmail.com

1. Introduction

Origins of classical music in the Indian subcontinent can be traced back to 1500 BC and is said to be described in Samaveda (Indian sacred books) [1]. It has progressed to the contemporary trends in Bollywood music and several regional folk forms. Indian classical music (ICM) adheres to well-established musical rules and practices, allowing artists to add improvisation to the compositions as they see fit. Music theory and understanding of musical ideas of ICM are passed from a Guru (teacher) to his/her Shishyas (students) in an oral tradition, and unorthodox ways unique only to a Guru [1]. Sargam and western musical staff notation are often used for educational, recreational reasons, and music documentation. However, no standard written notation for Indian classical music (ICM) exists. The lead artist discusses the upcoming performance that he/she is going to perform with accompanying artists utilizing concepts like Tala (Rhythm) and Raag[1] [2].

Musical scales in western classical music are fixed, and compositions are based on a fixed key signature. Contrary to that, musical scales in ICM are dynamic and change based on the underlying Tonic selected by the lead artist. A Saptak (Octave) splits into 22 microtones (shown in Table 1), each named a particular Shruti [3]. The first Shruti, Sa (called Shadja), is the Tonic of the music composition in ICM; Other Shrutis adopt their respective placements according to the Shadja. This understanding provides ICM musicians with much freedom. Shadja is reinforced in performances for reference and played on an instrument such as the Tabla or Tanpura. The Shadja to all other Shruti ratio inside

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[1] Raag is the melodic structure for classical Indian music, with grammar such as Pakad (characteristic phrase), Vadi (king note), and Samvadi (queen note), etc.

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Table 1
Frequency ratios of 22 Shrutis.

No.	Shruti mame	Symbol	Ratio	No.	Shruti name	Symbol	Ratio
1.	Shadja	S	1/1	12.	Teevra Madhyam	m1	45/32
2.	Ati Komal Rishabh	r1	256/243	13.	Teevratam Madhyam	m2	729/512
3.	Komal Rishabh	r2	16/15	14.	Pancham	P	3/2
4.	Shuddha Rishabh	R1	10/9	15.	Ati Komal Dhaivat	d1	128/81
5.	Teevra Rishabh	R2	9/8	16.	Komal Dhaivat	d2	8/5
6.	Ati Komal Gandhar	g1	32/27	17.	Shuddha Dhaivat	D1	5/3
7.	Komal Gandhar	g2	6/5	18.	Teevra Dhaivat	D2	27/16
8.	Shuddha Gandhar	G1	5/4	19.	Ati Komal Nishad	n1	16/9
9.	Teevra Gandhar	G2	81/64	20.	Komal Nishad	n2	9/5
10.	Shuddha Madhyam	M1	4/3	21.	Shuddha Nishad	N1	15/8
11.	Ek shruti Madhyam	M2	27/20	22.	Teevra Nishad	N2	243/128

an octave can be gathered from Table 1. For example, the ratio of the 14th Shruti (P) and Sa is 3/2. So, if Sa is 100 hertz, P would be 150 hertz. Seven Shrutis out of 22 are selected based on a given Raag and become seven Swaras (also known as Sargam) of that Raag. The Komal (flat) and Teevra (sharp) variations of these seven Swaras are as labeled $\{S, r, R, g, G, M, m, P, d, D, n, N\}$, which resembles the Western music notation of notes $\{C, C_\#, D, D_\#, E, F, F_\#, G, G_\#, A, A_\#, B\}$.

Further, as mentioned earlier, the Raag grammar gives the information on which Shrutis the performer can play. This arrangement is opposite to the western equal tempered scale (often used), in which the ratio of nearby halftones is constant and equal to $2^{\frac{1}{12}}$ — an octave is divided into 12 semitones [4]. Some feature extraction procedures can substitute with alternate techniques, such as STFT with CQT transform. More smoothing methods can be applied after the feature extraction procedure to improve the representation's temporal and tonality stability. The scientific community has made significant advancements in evaluating western music [5], and researchers have created many methodologies and technologies [6–9]. However, scientific contributions to Indian music remain limited [2] to comprehend its complexity correctly. For this purpose, we provide the code implementation of Swaragram [10] that:

1. Loads the music file given in standard audio formats and extracts the Tonic of the loaded file.
2. Implements the Swaragram feature and its underlying components.
3. Applies standard normalizations over the extracted feature.
4. and provides visualization of the extracted features for visual understanding.

2. Functionalities and key features

The Swaragram library is implemented in the Python programming language and published under the GNU-GPLv3 license on the GitHub.[2] Implementation is divided into two core sub-modules (`core` and `utils`) and uses good practice to keep them manageable. The `core` sub-module contains a collection of the required Swaragram functions and provides the help and wrapper functions with the `utils` sub-module. The Python functions and descriptions are described in Table 2. Users can acquire complete information on the default parameters and the API on Github, and the default values can be modified for desired parameter effects.

3. Impact overview

The Swaragram toolbox implements the Swaragram feature with supporting functionality specifically to be used for Indian music. Contrary to Chromagram, a widely used feature technique for spectrum analyses that analyzes the music's harmonic character and is used extensively for different music information retrieval purposes [11–16]. However, the fluid character of Swaras (notes) in ICM cannot be captured. It is because the frequencies of the Swaras alter according to the Tonic of music selected by the performer. In order to analyze ICM, we need a flexible and fluid Chromagram-like technique based on the Tonic (Shadja) and microtones (Shrutis). This is achieved by Swaragram.

Audio processing and the automation of Indian music recordings for Raag classification/identification is a significant contribution to retrieving music in the Indian context. Most Raag recognition techniques use pitch-class distributions, while others are modeled on vector space and melodic surfaces [17–19]. Chromagram is commonly used in Western music applications for harmonic analysis (particularly melodies) of audio recordings. The efficacy of the Chroma feature in ICM is insufficient in our observation. We have used the Dunya data set [20] to conduct a small experiment for Raag recognition. Extensive experiments are conducted in the following publications against several features.

1. Y. Singh and A. Biswas, "Robustness of musical features on deep learning models for music genre classification", Expert Systems with Applications, vol. 199, p. 116879, 2022 [21].
2. Y. Singh and A. Biswas, "Swaragram based residual neural architecture for raag identification in indian classical music", in 2021 12th International Conference on Computing Communication and Networking Technologies (ICCCNT), pp. 1–6, IEEE, 2021 [10].
3. Y. Singh, R. Kumar, and A. Biswas, "Swaragram: Shruti-based chromagram for indian classical music", in Advances in Speech and Music Technology (A. Biswas, E. Wennekes, T.-P. Hong, and A. Wieczorkowska, eds.), (Singapore), pp. 109–118, Springer Singapore, 2021 [22].

The boost in performance against the chromagram feature, which is also an abstract harmonic music feature, is due to the design of the Swaragram, which considers the Tonic and Shruti while computing the dynamic scale for a given recording. Further distinguishing these two features in terms of visual representations in Fig. 1. The figure displays the plots of the Chromagram and Swaragram of a sample ICM performance. The audio sample was a vocal performance of Raag Bhoopali tuned with a Tanpura tuned to $A_\#4 = 440$ hertz. The upper plot in Fig. 1 displays the Chromagram for chosen ICM music sample. Almost anywhere, we observe several light areas. These weakly lit areas are the appearance of notes at any given time interval of corresponding notes in the vertical axis.

The grammar of Raag Bhoopali [22] indicates that some Swaras are prohibited. In the Chromagram plot, we see almost all the luminous areas of the forbidden Swaras (around A and E, if we take Swaras corresponding to western notation for equal tempered scale). Contrary, when we observe the Swaragram plot of the given ICM music sample in the lower plot of Fig. 1, the M and N Swaras, forbidden in this Raag, are not present in the representation. It proves the validity of the Swaragram and its usefulness. Raag Bhoopali can only be performed using Shudha Swaras and the inclusion of Komal Swaras in our plots. It is caused by the use of Alankars (ornamentation) such as Meend (gliding) by the musicians in the Chalan (progression of notes) of a given. The Swaragram may be further processed to tolerate tonal and time differences, and others are urged to do so.

[2] https://www.github.com/yeshwantsingh/swaragram.

Table 2
Description of the API of the Swaragram library.

Function	Description
`utils submodule`	
`computeRefPitch`	It computes the MIDI pitch value corresponding to the Shadja (Tonic) of the give music piece.
`computeSTFT`	It computes the Short-Term Fourier Transform (STFT) of the given music piece.
`getTonic`	It computes the Shadja (Tonic) of the given music piece.
`loadAudio`	It loads the audio sample at a given path.
`logCompression`	It compute the Log compression of the given input.
`core submodule`	
`swaraRange`	It computes the higher and lower frequencies of a MIDI pitch value based on Shurti scale.
`poolMask`	It computes the set of frequency indices that are assigned to a given pitch.
`computeSwaragram`	It computes the Swaragram of a given STFT Spectrogram and Tonic value.
`logFreqSpec`	It computes a log-scale frequency Spectrogram.
`plotSwaragram`	It help to visualize the Swaragram.

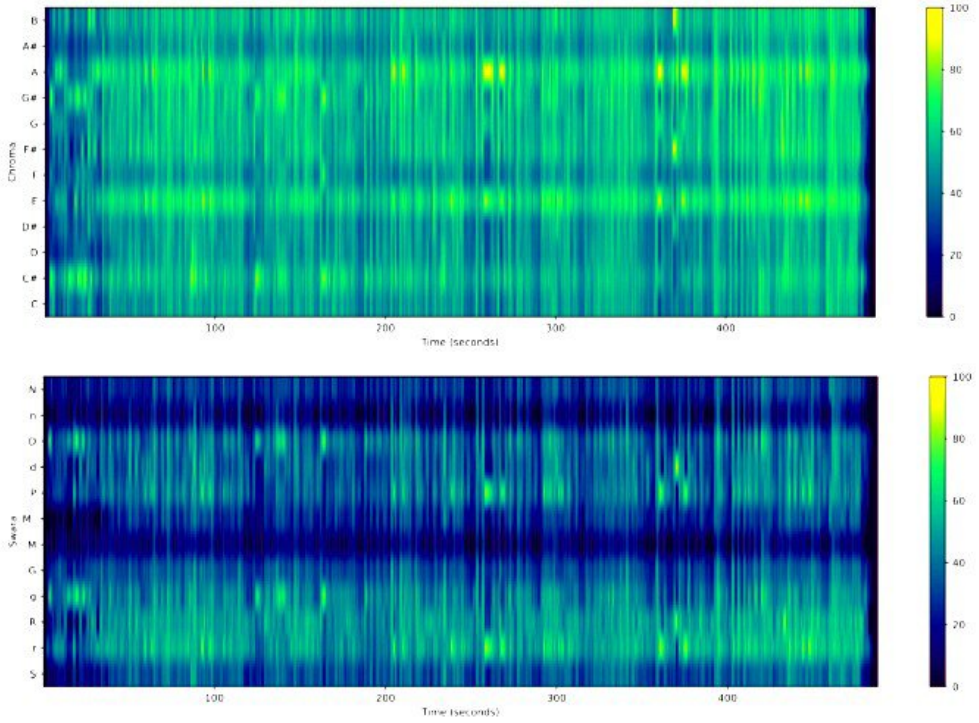


Fig. 1. Chromagram and Swaragram plot of ICM audio sample.

4. Conclusions and further development

We have presented the Swaragram Python library, which extracts Swaragram, a feature inspired by the chromagram but focuses on Indian music. It is implemented with Python standards and also includes utility functions. The Python library is freely licensed to allow other researchers focused on Indian music to engage in MIR. Furthermore, we applied this technique in classical Indian music to identify Raag. Others can develop applications using the Swaragram Python library and can contribute to adding smoothing techniques such as normalization to tolerate noise and loudness variations better or by adding new features. For future development, the Swaragram feature and toolbox can be updated to perform Artist Identification and Song matching in a large corpus. Another added functionality can be incorporating the library with tonic recognition functionality.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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